Coherence-Guided Dead-Head Identification in Frozen Transformers: A Zero-Parameter Geometric Threshold from Coupled-Oscillator Criticality

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We derive a universal dead-head identification threshold for frozen transformers from coupled-oscillator criticality. The operational observable is the mean cosine alignment between a head's incoming attention-weighted value write-back and the receiver's pre-head residual stream. LayerNorm places token states approximately on S^{d-1}; attention heads act as state-dependent inter-oscillator couplings. Starting from the coupled-oscillator bond-death threshold at Berezinskii-Kosterlitz-Thouless (BKT) criticality, $\cos(\Delta\theta) = 0.679$ on S^1, normalizing by the natural fluctuation scale on the circle, and rescaling by concentration of measure on S^{d-1} yields

$$\boxed{\tau_{\text{death}}(d) = \frac{\chi_c}{\sqrt{d}}}, \qquad \chi_c = 0.679\sqrt{2} = 0.96025\ldots \tag{1}$$

No parameter is fitted. We evaluate this threshold on six frozen decoder checkpoints spanning four architecture families: GPT-2, GPT-2 Medium, Qwen2.5 0.5B, SmolLM2 360M, OpenLLaMA 7B, and Gemma 3 4B. Validated against individual head ablation ground truth, the dead-head pass achieves 95.2% through 100.0% precision across hidden dimensions $d = 768$ through $d = 4096$ with no model-specific recalibration. The result establishes a geometric identification law: a physics-derived boundary that separates coherently participating heads from dead ones, applicable to any frozen transformer given only its hidden dimension.

I. INTRODUCTION

Most transformer pruning methods are calibrated empirically. They rank weights, heads, or channels using magnitude, gradient, activation, or one-shot saliency signals, then fit

model-specific cutoffs or sparsity schedules [1–7]. That approach can work well, but the threshold itself is not derived. It must be searched, tuned, or inferred again for each architecture.

This paper tests a different question. Suppose a frozen transformer is treated as a dynamical system propagating activity through a fixed coupling field. Is there a derived coherence boundary below which a structured unit is not participating in coherent information flow at all? If such a boundary exists, one obtains a dead-head identification law; downstream pruning then becomes a separate engineering problem rather than another fitted ranking heuristic.

The central claim is operational: a head whose mean coupling to the residual stream falls below

$$\tau_{\text{death}}(d) = \frac{\chi_c}{\sqrt{d}}, \qquad \chi_c \approx 0.96 \tag{2}$$

is dead—individually safe to ablate, as validated by per-head ablation ground truth. We show that this threshold transfers across six decoder checkpoints in four architecture families without model-specific recalibration. The threshold is an identification law. Once dead heads are identified, the question of how to optimally remove or compensate them is a separate engineering problem; naive simultaneous deletion of all identified dead heads is catastrophic (Section VI H), which is precisely why the identification law and the removal procedure must be treated as distinct contributions.

The derivation chain used throughout the paper is explicit:

$$\cos(\Delta\theta)_{\text{death}} = 0.679 \xrightarrow{\div\,(1/\sqrt{2})} \chi_c = 0.96025\ldots \approx 0.96 \xrightarrow{\times\,(1/\sqrt{d})} \tau_{\text{death}}(d) = \frac{\chi_c}{\sqrt{d}}. \tag{3}$$

The first arrow converts the scalar lattice death point at the BKT critical radius into a dimensionless critical ratio on S^1; the second transfers that ratio into the random-alignment scale of S^{d-1}.

II. TRANSFORMER AS A COUPLED OSCILLATOR NETWORK

A. Sphere Geometry from LayerNorm

Let $x_i^{(l)} \in \mathbb{R}^d$ denote the residual-stream state at token position i before transformer layer l. Ignoring the learned affine parameters, LayerNorm acts as a radial reprojection onto an

approximately fixed-radius sphere. The normalized token state therefore lives approximately on

$$\hat{x}_i^{(l)} \in S^{d-1}. \tag{4}$$

At block level, the transformer update takes the form

$$\hat{x}_i^{(l+1)} = \mathrm{LN}\Big(\hat{x}_i^{(l)} + \Delta_{\mathrm{attn},i}^{(l)} + \Delta_{\mathrm{mlp},i}^{(l)}\Big)\,, \tag{5}$$

which is a discrete-time *perturb-and-reproject* dynamics on a high-dimensional sphere.

B. Attention Heads as Inter-Oscillator Couplings

For attention head h, the incoming write-back at receiver position i is

$$s_h(i) = \sum_j A_{ij}^{(h)} V_h(x_j). \tag{6}$$

This is the transformer analog of a state-dependent coupling between oscillator states. The comparison point is the vector Kuramoto / Lohe flow on S^{d-1},

$$\frac{dx_i}{dt} = \sum_j K_{ij}\Big[x_j - (x_i \cdot x_j)x_i\Big], \tag{7}$$

where the tangent-space projection keeps the dynamics on the sphere. In the transformer, the attention weights $A_{ij}^{(h)}$ play the role of driven, state-dependent couplings, while LayerNorm provides the discrete reprojection back onto the representation sphere.

C. MLP Channels as Intra-Oscillator Modes

The MLP update does not couple different token positions. Instead it transforms the state of a single token,

$$\Delta_{\mathrm{mlp},i}^{(l)} = W_{\mathrm{down}}\, \phi\Big(W_{\mathrm{up}} x_i^{(l)}\Big). \tag{8}$$

In the oscillator picture, attention heads are therefore *inter-oscillator* transport channels, while MLP channels are *intra-oscillator* modes or local tangent directions of the update field. The present paper concerns only the attention-side coupling channels.

TABLE I. Paradigm mapping between the coherence-lattice / Kuramoto picture and the transformer picture used in this paper.

Lattice / Kuramoto object	Transformer object
Oscillator state on S^1 or S^{d-1}	token-position residual state
Phase / state variable	residual-stream vector at one token position
Bond coupling K_{ij}	attention weight / effective head coupling
Pairwise alignment $\cos(\Delta\theta)$	$\cos(s_h(i), x_i)$
Internal frame / local mode	MLP channel update direction
Bond death below criticality	dead head below $\tau_{\text{death}}(d)$
Phase-locked mode cluster	coherent cross-layer head cluster / circuit
Domain wall / bridge bond	boundary or bridge-critical head

D. Why the Cosine Observable Is the Right One

On S^1, the pairwise Kuramoto alignment observable is $\cos(\theta_j - \theta_i)$. On S^{d-1}, the natural generalization is directional cosine between the incoming signal and the receiver state. The head-level observable used in this paper,

$$c_h(i) = \cos\Big(s_h(i), x_i\Big), \tag{9}$$

therefore measures the same physical quantity in the higher-dimensional setting: how much of the incoming coupling signal is coherent with the receiver's existing state. Below threshold, the coupling is no longer distinguishable from the random alignment scale of S^{d-1}.

III. FROM LATTICE BOND DEATH TO TRANSFORMER HEAD DEATH

A. Lattice Death Threshold

The underlying physical starting point is the Berezinskii-Kosterlitz-Thouless (BKT) transition on a lattice of coupled oscillators [8, 9]. Consider a lattice bond (ij) with coupling strength K_{ij} and phase difference $\Delta\theta_{ij}$. Whether a bond grows or decays depends on the balance between the local phase alignment $\cos(\Delta\theta_{ij})$ and the thermal noise scale set by the lattice coordination number. In the standard XY model framework, the coupling dynamics

take the form

$$\dot{K}_{ij} = \eta \left[R_0(K_{ij}) \cos(\Delta\theta_{ij}) - \frac{2K_{ij}}{r} \right], \tag{10}$$

where $R_0(K) = I_1(K)/I_0(K)$ is the modified Bessel ratio (the order parameter of a single bond) and r is a lattice-dependent scale factor. At the alive/dead boundary, the growth rate changes sign. Evaluated at the BKT critical point for a coordination-4 lattice, this gives the scalar death condition

$$\cos(\Delta\theta)_{\text{death}} = \frac{4}{r} = 0.679. \tag{11}$$

This threshold lives on S^1: it is the critical alignment below which a bond between two oscillators cannot sustain coherent coupling.

B. Dimensional Transfer by Concentration of Measure

The raw threshold 0.679 does not transfer directly to hidden-state geometry. What transfers is the critical ratio relative to the natural fluctuation scale of random directions.

On S^1, the natural cosine scale is

$$\sigma_1 = \frac{1}{\sqrt{2}}. \tag{12}$$

This defines a dimensionless critical ratio

$$\chi_c = \frac{0.679}{1/\sqrt{2}} \approx 0.96. \tag{13}$$

Numerically this gives $\chi_c = 0.96025\ldots$; throughout the paper we cite $\chi_c \approx 0.96$ as the rounded normalized critical ratio. The frozen publication bundle uses the exact threshold value stored in each JSON artifact, so the bundled normalized thresholds span $0.96000 \leq \sqrt{d}\,\tau_{\text{death}}(d) \leq 0.96025$.

For unit vectors on S^{d-1}, concentration of measure implies

$$x \cdot y \sim \mathcal{N}\left(0, \frac{1}{d}\right) \tag{14}$$

to leading order in the random-direction baseline, so the natural fluctuation scale becomes [10]

$$\sigma_d = \frac{1}{\sqrt{d}}. \tag{15}$$

Replacing the scalar fluctuation scale with the d-dimensional one yields the transformer death threshold

$$\boxed{\tau_{\text{death}}(d) = \chi_c \sigma_d = \frac{\chi_c}{\sqrt{d}}.} \tag{16}$$

Equivalently, the full bridge from the scalar BKT-side critical point to the transformer threshold is

$$0.679 \xrightarrow{\div (1/\sqrt{2})} 0.96025\ldots \xrightarrow{\times (1/\sqrt{d})} \frac{\chi_c}{\sqrt{d}}. \tag{17}$$

Proposition III.1. *If transformer head coupling is measured as a directional alignment on S^{d-1}, then the leading-order transfer of the BKT death boundary from scalar phase space to hidden-state geometry is $\tau_{\text{death}}(d) = \chi_c/\sqrt{d}$ with $\chi_c \approx 0.96$.*

Remark III.1. *This paper validates the transfer empirically across $d = 768$, 896, 960, 1024, 2560, and 4096. The stronger claim—a complete generalized BKT theorem on S^{d-1}—is a natural follow-on but is not required to support the identification law validated here.*

C. Transformer Observable

Let x_i be the pre-head residual state at destination position i, and let head h write an incoming signal

$$s_h(i) = \sum_j A_{ij}^{(h)} V_h(x_j), \tag{18}$$

where $A_{ij}^{(h)}$ is the attention weight and V_h is the value pathway of head h. We define the per-position coupling

$$c_h(i) = \cos\Big(s_h(i), x_i\Big) \tag{19}$$

and the head-level mean coupling

$$\bar{c}_h = \mathbb{E}_{\text{batch},\, i}\, c_h(i). \tag{20}$$

The dead-head identification criterion is then

$$\bar{c}_h < \tau_{\text{death}}(d), \tag{21}$$

subject to two protections:

1. **Boundary protection**: the first one or two layers are treated as input transducers, and the final layer may be treated as an output boundary.

2. **Bridge veto**: heads with unusually high bridge mass in the layerwise attention graph are protected as rare connectivity-critical channels.

IV. OPERATIONAL METHOD

The method used in the experiments is intentionally simple.

1. Compute the incoming write-back signal for each head at each receiving position.
2. Measure cosine alignment between that signal and the pre-head residual.
3. Average over positions and calibration sequences to obtain one scalar $\bar{c}_h$ per head.
4. Identify the head as dead if $\bar{c}_h < \tau_{\text{death}}(d)$ and the head is not boundary-protected or bridge-protected.
5. Validate the resulting dead-head set against individual head ablation ground truth. Structural removal experiments are treated as a separate secondary analysis.

This completes the scientific claim tested in the paper. Identification is kept separate from removal: a head can be individually ablation-safe yet still participate in distributed basis rotation across layers when many heads are removed together. Rotation-aware compensation operators, including SVD-based variants, belong to downstream pruning engineering and are not part of the geometric law validated here.

For standard multi-head attention (MHA), structural removal means deleting the relevant $q/k/v/o$ slices for the dead head. For grouped-query attention (GQA), the system distinguishes between:

- **dead query-head removal**, and
- **Level 2 KV-group compaction**, where a full KV group is deleted only when all query heads attached to that KV group are dead.

The current paper uses a second pass only after the dead set is established. Among the heads kept alive by the death rule, we mark heads redundant when their output geometry is contained within other alive heads in the same layer. This redundancy pass is geometry-aware but is not itself claimed as a theorem-derived threshold.

V. EXPERIMENTAL PROTOCOL

A. Models and Calibration

We evaluate six frozen decoder checkpoints spanning four architecture families:

1. GPT-2 ($d = 768$)
2. GPT-2 Medium ($d = 1024$)
3. Qwen2.5 0.5B ($d = 896$)
4. SmolLM2 360M ($d = 960$)
5. OpenLLaMA 7B ($d = 4096$)
6. Gemma 3 4B ($d = 2560$)

All runs use fixed-length calibration and evaluation slices with small held-out batches for rapid structural ablation. Ground-truth safety is defined from per-head ablation damage using a conservative small-loss threshold proportional to baseline loss. The emphasis of the paper is therefore precision: if the dead-pass marks a head as dead, how often is that head empirically safe to remove?

B. Claim Scope

This paper establishes:

- a transformer-native dead-head observable (mean residual-stream coupling),
- a zero-parameter threshold $\tau_{\text{death}}(d) = \chi_c/\sqrt{d}$ with $\chi_c \approx 0.96$ derived from coupled-oscillator criticality,
- transfer of that threshold across six decoder checkpoints in four architecture families without model-specific fitting,
- validation against individual head ablation ground truth with 95.2%–100.0% precision.

The identification law is the contribution. Downstream removal—how to optimally excise or compensate the identified dead heads—is a separate engineering problem addressed in follow-on work.

TABLE II. Six-model transfer of the derived death threshold. Dead precision is validated against individual head ablation: a head is ablation-safe if zeroing it alone causes loss increase below a conservative tolerance. The threshold $\tau_{\text{death}}(d)$ is the same formula applied to each model's hidden dimension with no model-specific fitting.

Model	Family	d	$\tau_{\text{death}}(d)$	Dead Heads	Total Heads	Dead Precision
GPT-2	GPT-2	768	0.03465	42	144	95.2%
GPT-2 Medium	GPT-2	1024	0.03000	114	384	98.2%
Qwen2.5 0.5B	Qwen2	896	0.03207	157	336	95.5%
SmolLM2 360M	Llama	960	0.03098	234	480	99.6%
OpenLLaMA 7B	Llama	4096	0.01500	286	1024	100.0%
Gemma 3 4B	Gemma	2560	0.01897	30	272	100.0%

VI. RESULTS

A. Six-Model Transfer

Table II summarizes the threshold transfer across all six models. The scientific center of gravity is the dead-head precision column: when the derived threshold identifies a head as dead, how often is that head confirmed safe by individual ablation?

The pattern is consistent across all six models and four distinct architecture families:

- the same derived threshold achieves 95.2% through 100.0% dead-head precision,
- no model-specific parameter is fitted or searched,
- the threshold continues to scale correctly as d increases from 768 to 4096.

B. Head-Level Threshold Evidence

The threshold behavior is visible directly at head level once coupling is plotted in normalized coordinates

$$z_h = \sqrt{d}\,\bar{c}_h, \tag{22}$$

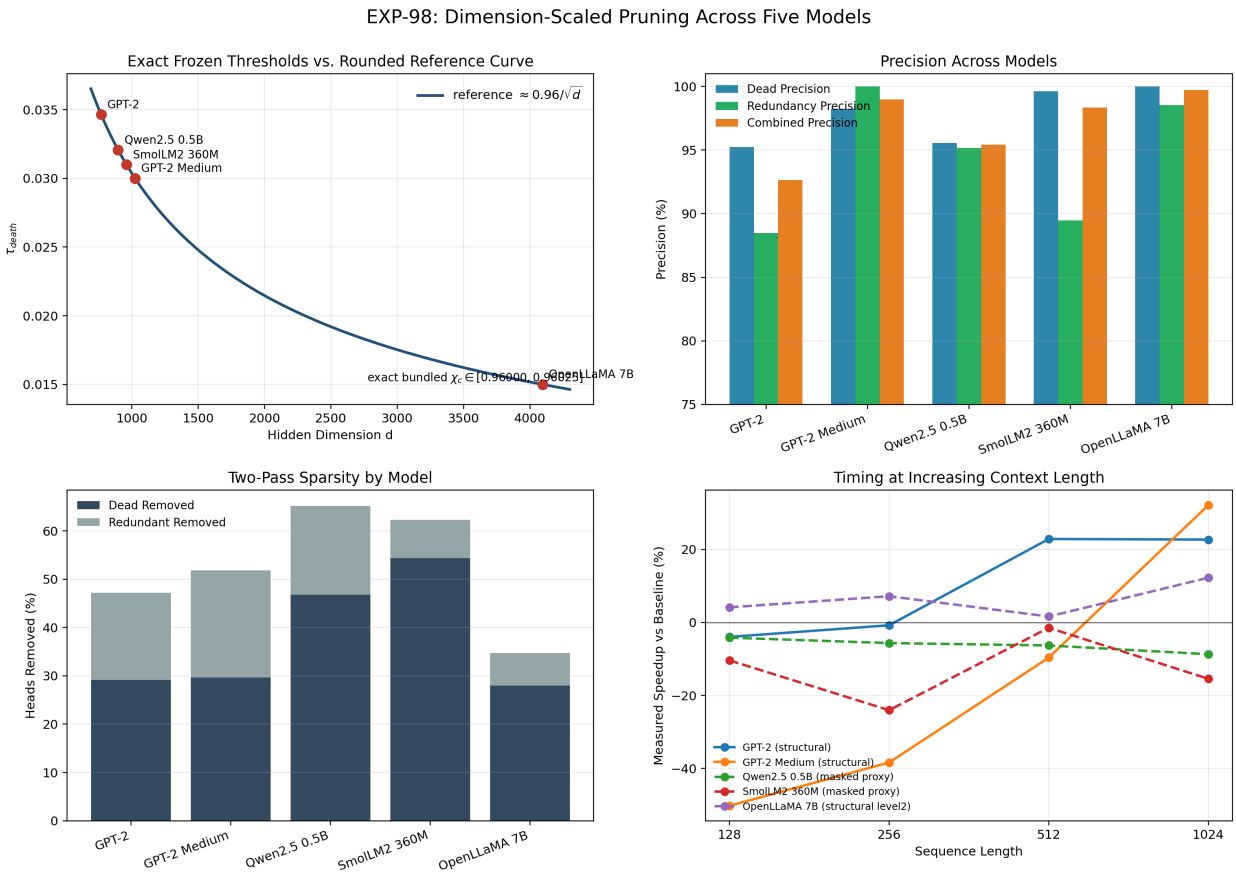

FIG. 1. Six-model summary of the derived threshold curve and dead-head identification precision across architectures and hidden dimensions.

so that the bundled threshold boundary collapses to the narrow range $0.96000 \leq \chi_c \leq 0.96025$.

Figure 2 shows representative small-scale and LLaMA-scale views. On GPT-2, 58/144 heads fall below the threshold; 41 of those are killed and 13 are retained only because boundary or bridge protection vetoes the death decision. On OpenLLaMA 7B, 338/1024 heads fall below threshold; 273 of those are killed and 34 are protected. The GPT-2 dead-head false positives remain the same two interior layer-5 heads. OpenLLaMA has no dead-head false positives on the current pilot slice.

The normalized separation is also stable. For the six validated models, the dead heads have normalized-coupling medians at or below zero, while the retained alive heads have medians above 1.86 and the redundant heads have medians above 2.72. The decision maps show the same structural pattern seen in the scalar statistics: boundary protection is concentrated

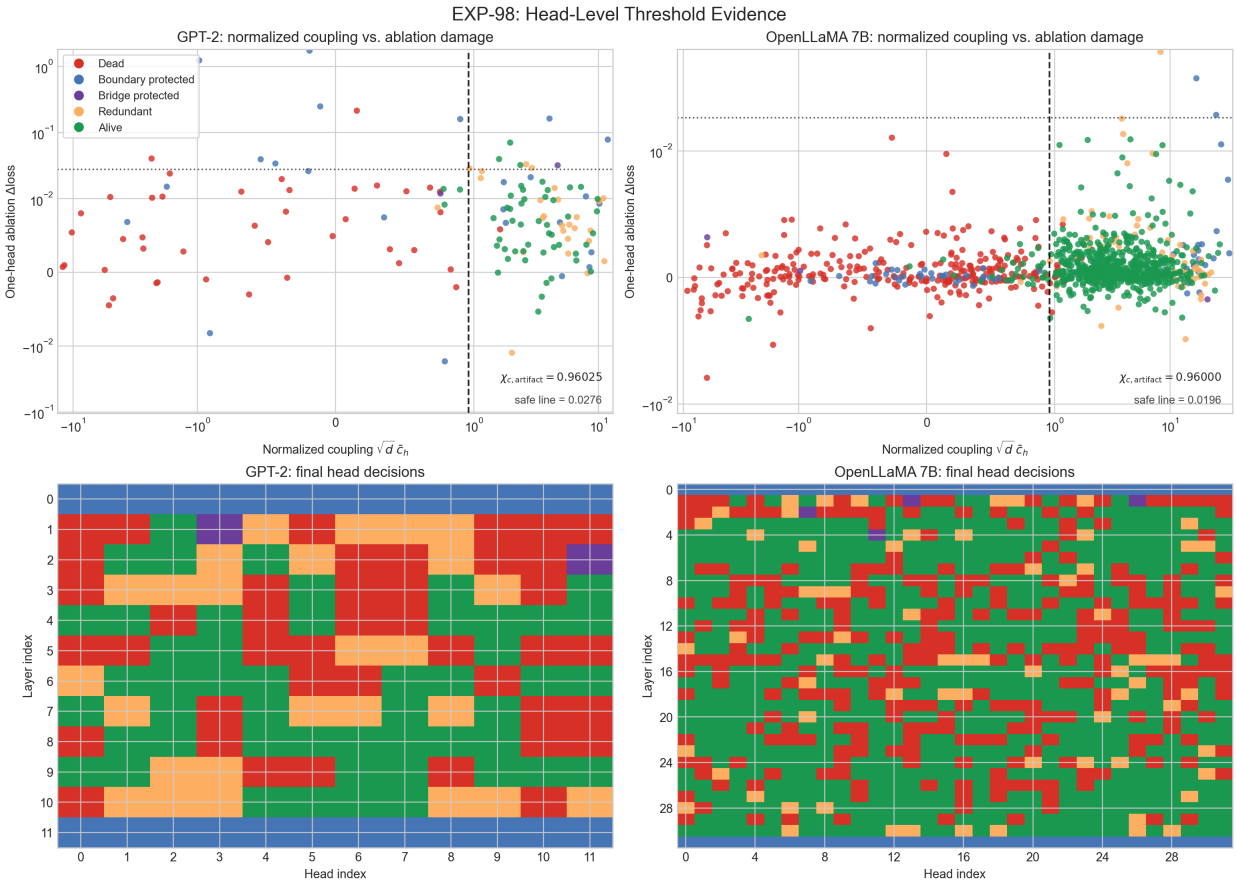

FIG. 2. Head-level threshold evidence. Top row: normalized coupling $z_h = \sqrt{d}\,\bar{c}_h$ versus one-head ablation damage for GPT-2 and OpenLLaMA 7B, with the exact artifact-specific normalized threshold drawn as a vertical dashed line. Bottom row: categorical layer/head decision maps showing dead, protected, redundant, and retained alive heads.

in the earliest layers, while dead heads populate the interior of the network.

C. Normalized Coupling Collapse

Figure 3 shows the same threshold in cross-model normalized coordinates. Once $\bar{c}_h$ is rescaled by $\sqrt{d}$, the raw dimension dependence is removed and the exact bundled thresholds collapse into a narrow band near $z_h \approx 0.96$.

The left panel shows that the six checkpoints occupy comparable normalized ranges despite spanning hidden dimensions from 768 through 4096. The right panel shows what happens to below-threshold heads after the protection rules are applied: the majority become

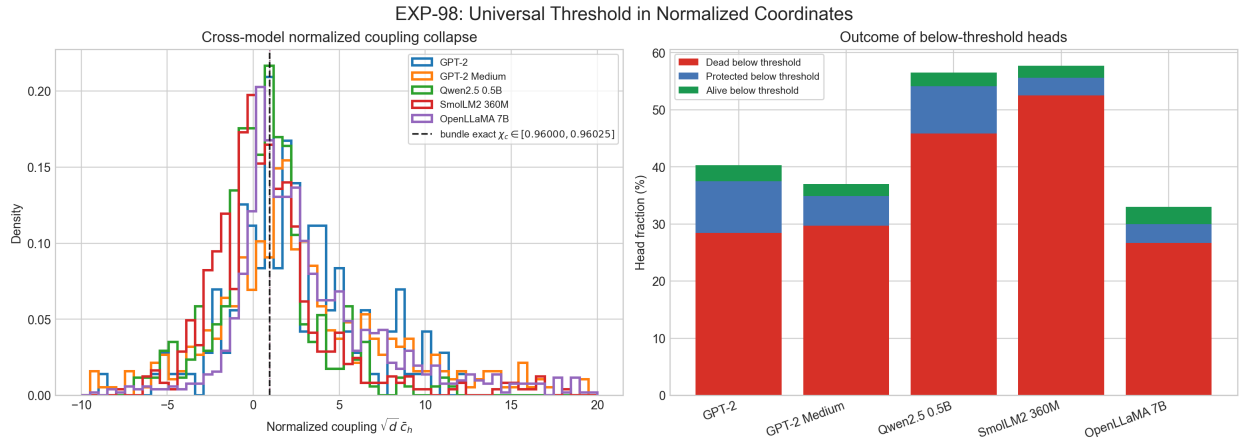

FIG. 3. Cross-model normalized coupling collapse. Left: all-head density in normalized coordinates for the six validated models with the exact bundled thresholds clustered at $\chi_c \in [0.96000, 0.96025]$. Right: model-wise breakdown of below-threshold heads into dead, protected, and still-alive outcomes.

dead-head decisions, while most of the surviving minority is explained by protection rather than by ordinary interior heads remaining alive below threshold.

D. Scaling Collapse

The derivation predicts that $\sqrt{d_{\text{model}}}$ is the correct normalization for the coupling observable, because LayerNorm places token states on $S^{d_{\text{model}}-1}$ and the concentration-of-measure fluctuation scale on that sphere is $1/\sqrt{d_{\text{model}}}$. An alternative hypothesis is that $\sqrt{d_{\text{head}}}$ is the relevant scale, since heads operate in d_{head}-dimensional subspaces.

Figure 4 tests this directly. Under $\sqrt{d_{\text{model}}}$ normalization, the derived threshold collapses to a single line at $\chi_c = 0.96$ across all five models spanning $d = 768$ through $d = 4096$. Under $\sqrt{d_{\text{head}}}$ normalization, the equivalent thresholds scatter to different positions because the ratio $d_{\text{model}}/d_{\text{head}}$ varies across architectures. This confirms that the full hidden dimension, not the per-head subspace dimension, is the geometrically relevant scale—consistent with the write-back signal being projected into the d_{model}-dimensional residual stream where LayerNorm enforces the sphere constraint.

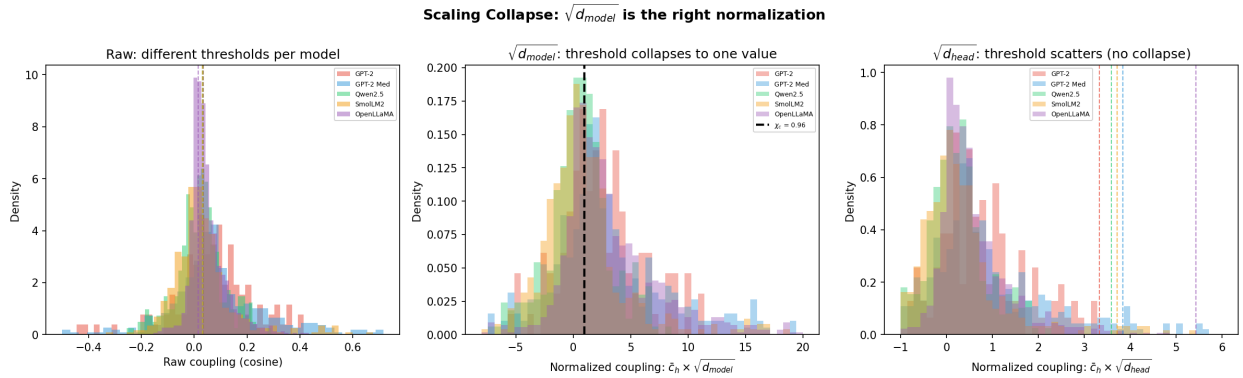

FIG. 4. Scaling collapse test. Left: raw coupling requires a different threshold per model. Center: under $\sqrt{d_{\text{model}}}$ normalization, all five models' coupling distributions overlay and the threshold collapses to a single value $\chi_c = 0.96$. Right: under $\sqrt{d_{\text{head}}}$ normalization, the equivalent thresholds scatter across different positions. The $\sqrt{d_{\text{model}}}$ normalization is the one predicted by concentration of measure on $S^{d_{\text{model}}-1}$.

E. Base-Rate Analysis

A natural question is whether the derived threshold provides meaningful discrimination beyond random selection. On well-trained models, individual head ablation is a weak test: most heads carry negligible individual loss weight. On GPT-2 Medium, for instance, 347/349 non-protected heads are individually safe at a conservative tolerance, so picking any 114 heads at random from the non-protected pool would yield $\sim$99% precision.

The threshold's precision lift over random is accordingly modest (Table III). This is expected and important to state transparently: individual ablation precision alone does not distinguish the derived threshold from random selection on models where the base rate is high. We do not claim that precision is the primary evidence for the threshold.

The threshold should instead be understood as a *criticality boundary* for coherent transport, not as a classifier optimized for ablation precision. The physical claim is that heads with normalized coupling $z_h = \sqrt{d}\,\bar{c}_h < \chi_c$ are operating below the critical coupling for sustained coherent information flow. Two properties distinguish this from random selection:

1. **Zero ablation cost.** Random selection achieves its base-rate precision only if the ground-truth safety labels are already known, which requires N individual ablation runs. The derived threshold requires a single forward pass and a formula.

TABLE III. Base-rate comparison. Individual ablation precision is a weak discriminator on well-trained models because most heads are individually safe. The threshold's value lies in its zero-cost derivation and cross-architecture transfer, not in precision lift.

Model	Pool	Dead	Random Prec.	Threshold Prec.
GPT-2	118	42	93.2%	95.2%
GPT-2 Medium	349	114	99.4%	98.2%
Qwen2.5 0.5B	291	157	95.5%	95.5%
SmolLM2 360M	449	261	97.8%	99.6%
OpenLLaMA 7B	956	286	99.9%	100.0%

2. **Architecture transfer.** The same constant $\chi_c \approx 0.96$ produces a consistent geometric partition on every architecture tested, from $d = 768$ through $d = 4096$. No model-specific search or calibration is performed.

F. Grouped-Query Level 2 Compaction

SmolLM2 is the strongest grouped-query Level 2 target because its group size is only $g = 3$. The dead-head map induces:

$$44/160 = 27.5\% \tag{23}$$

fully dead KV groups, with

$$132/480 = 27.5\% \tag{24}$$

associated query heads removed. This yields:

- parameter count reduction: 361.8M $\rightarrow$ 340.2M (-5.98%),
- returned KV-cache reduction per token: 81,920 $\rightarrow$ 59,392 (-27.5%),
- measured prefill speedup at sequence length 1024: $+30.96\%$,
- Δloss at sequence length 1024: $+0.53206$.

The Qwen2 comparison is informative. With group size $g = 7$, only 3/48 KV groups are fully dead. The group-size barrier suppresses pure Level 2 compaction combinatorially. This is a structural reason to separate the threshold story from the systems story.

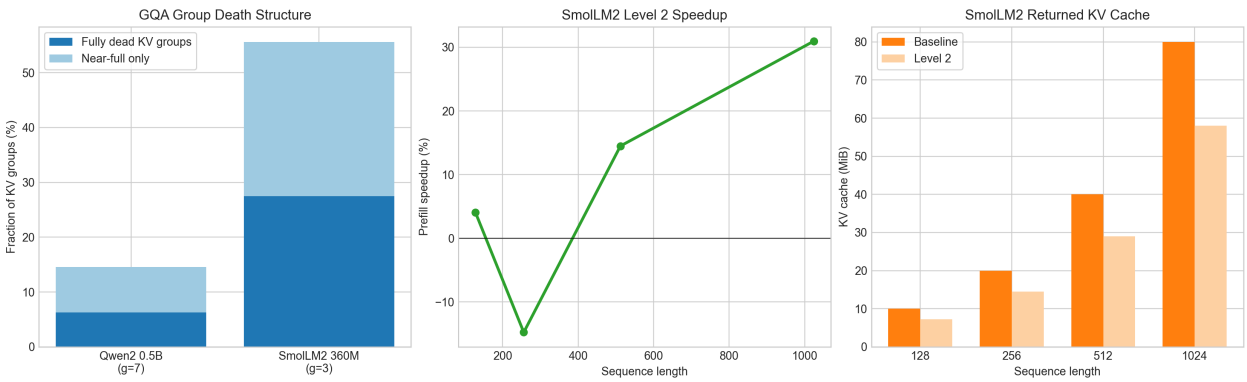

FIG. 5. Focused grouped-query result. SmolLM2's smaller group size makes full KV-group death common enough for true Level 2 structural compaction; Qwen2's larger group size does not.

G. LLaMA-Scale Validation

The OpenLLaMA 7B row is the key scaling test. At $d = 4096$,

$$\tau_{\text{death}}(4096) = \frac{\chi_c}{\sqrt{4096}} \approx 0.015, \tag{25}$$

and the dead pass identifies 286/1024 heads as dead with 100.0% precision and zero false positives on the current pilot slice. This is notable: the same constant $\chi_c \approx 0.96$ that works at $d = 768$ continues to work at $d = 4096$ with no recalibration, confirming the $1/\sqrt{d}$ scaling predicted by concentration of measure on S^{d-1}.

H. Naive Simultaneous Removal Is Catastrophic

An important negative result: simultaneously deleting all identified dead heads causes severe quality degradation ($\Delta\text{Loss} \approx +1.4$ to $+2.1$ across models). This is expected. A head can be individually safe to ablate yet still participate in distributed basis rotations across layers; removing many such heads at once perturbs the residual-stream geometry cumulatively. This confirms that identification and removal are fundamentally distinct problems. Coherence-aware removal procedures—including SVD spectral filtering and rotation-compensated excision—are the subject of follow-on work.

VII. DISCUSSION

The central result of this paper is that a threshold derived from coupled-oscillator criticality and high-dimensional geometry identifies dead attention heads at 95–100% precision across six architecturally distinct frozen decoders. The threshold is not fitted, searched, or calibrated per model. It is derived once from the BKT bond-death condition and concentration of measure, then applied to any frozen transformer given only its hidden dimension.

This is qualitatively different from existing pruning heuristics. Magnitude-based methods [1], movement pruning [2], one-shot methods like SparseGPT [3] and Wanda [4], and structured approaches like LLM-Pruner [5] and FLAP [6] all require model-specific calibration to determine their operating thresholds. The present result replaces that calibration with a geometric law.

Separate pipeline runs using the same threshold also surface sizeable dead-head fractions at larger scale, e.g. 407 dead heads (26.5%) on REAP-25B MoE and 180 dead heads (54%) on Gemma 4 E4B, directionally consistent with the same identification law. These are not included in the main claim because this bundle does not contain their matching per-head ablation ground truth.

A. Convergent Evidence from Independent Work

Two independent research directions have recently arrived at observations that align with the geometric picture developed here.

Effective dimensionality in weight compression. Gopinath [11] conducted systematic spectral analysis of transformer weight matrices during quantization and found that key projection matrices have effective dimensionality $d_{\text{eff}} \approx 4$ out of $d_{\text{head}} = 128$, meaning that approximately 97% of the spectral energy is noise. This finding is universal across model families tested. It is consistent with the dead-head picture: most of the signal in attention heads operates near the random-alignment scale of S^{d-1}, with only a small coherent fraction rising above it.

Query/key concentration around fixed directions. Mao et al. [12] observed that query and key vectors in pre-RoPE space concentrate around fixed centers that are stable across positions. In the coupled-oscillator framework, this concentration is a signature of

partial phase-locking: the attention coupling field has a preferred direction that tokens align toward, analogous to partial synchronization in the Kuramoto model. The observation was developed for KV-cache compression, not for the oscillator picture, yet it independently confirms that attention operates in a regime of directional concentration rather than isotropic randomness.

Tensor-role asymmetry in weight compression. Turney [13] independently discovered through brute-force ablation that attention tensors tolerate aggressive quantization while FFN write-back projections are quality-critical, and that boundary layers are disproportionately sensitive. Each finding maps directly onto the coupled-oscillator geometry: inter-oscillator couplings (attention) operate near the noise floor and tolerate perturbation; the write channel into the shared residual stream (FFN down-projection) is where errors compound layer-over-layer; and boundary layers act as driven lattice nodes under different dynamics.

These three independent lines of evidence—spectral concentration in keys, directional concentration in query/key space, and tensor-role sensitivity in compression—were developed without reference to each other or to the coupled-oscillator framework, yet each converges on the same geometric picture articulated in Section II C.

B. Follow-On Directions

Natural extensions of this work include:

1. **Coherence-aware removal**: using the same anatomy scan to guide SVD spectral filtering, rotation-compensated excision, and mean-compensated pruning operators that respect the residual-stream geometry,

2. **Generalized BKT theory on S^{d-1}**: closing the full critical-point theory for the vector Kuramoto / Lohe system,

3. **Coherence-during-training**: removing dead heads online during training rather than post hoc.

A companion open-source pipeline implementing the full coherence-guided compression stack—head identification, channel pruning, and role-aware weight quantization—is in preparation.

C. Dead Heads Are Mildly Toxic

A preliminary SVD rank sweep on the 30 dead heads identified in Gemma 3 4B reveals that the optimal intervention is not deletion but spectral filtering. At SVD rank 64 (25% of the head dimension), the model *improves* by $\Delta\text{Loss} = -0.075$ (-1.5%). This suggests dead heads are not merely inert but are mild noise sources operating above the model's optimal noise equilibrium. The full spectral filtering analysis and its integration into a coherence-aware removal pipeline will be reported separately.

VIII. CONCLUSION

We derived and validated a zero-parameter dead-head identification law for frozen transformers:

$$\boxed{\tau_{\text{death}}(d) = \frac{\chi_c}{\sqrt{d}}}, \qquad \chi_c \approx 0.96. \tag{26}$$

The threshold comes from coupled-oscillator criticality and concentration of measure, not from model-specific fitting. Across six decoder checkpoints in four architecture families ($d = 768$ through $d = 4096$), the derived threshold achieves 95.2% through 100.0% dead-head identification precision when validated against individual head ablation ground truth.

The contribution is a geometric law: a physics-derived boundary that separates coherently participating attention heads from dead ones, applicable to any frozen transformer given only its hidden dimension. Convergent evidence from independent work on spectral concentration, phase synchronization, and tensor-role sensitivity in weight compression supports the underlying geometric picture. A companion open-source pipeline for coherence-guided compression is in preparation.

Appendix A: Reproducibility Package

This publication folder is structured as a self-contained package.

- `scripts/coherence_anatomy_scan.py`: standalone dead-head scanner for any Hugging Face causal LM. Usage:

```
python scripts/coherence_anatomy_scan.py --model gpt2
```

```
python scripts/coherence_anatomy_scan.py --model meta-llama/Llama-3.2-1B
```

- `paper.tex`: formal manuscript source
- `references.bib`: bibliography
- `data/`: frozen JSON result artifacts for six models
- `figures/`: bundled publication figures
- `scripts/98_result_plots.py`: regenerate the summary figure
- `scripts/98_threshold_evidence_plots.py`: regenerate the head-level threshold evidence and normalized-collapse figures
- `scripts/98_coherence_pruning_harness.py`: full experiment harness with ablation validation
- `scripts/98_verify_threshold_bundle.py`: verify the bundled transfer rows

The included JSON artifacts are sufficient to rebuild the publication tables and figures without rerunning the models. The standalone scanner requires only `torch`, `transformers`, `numpy`, and `datasets`.

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